ENGINEERING ECONOMY

LECTURE 6

PROF.DR. YASEMİN ASU ÇIRPICI, MU, DEPT. OF ECONOMICS

Interest Rates That Vary Over Time

- Real-world interest rates for a corporation vary from year to year, depending upon the financial health of the corporation, its market sector, the national and international economies, forces of inflation, and many other elements.
- When *P*, *F*, and *A* values are calculated using a constant or average interest rate over the life of a project, rises and falls in *i* are neglected. If the variation in *i* is large, the equivalent values will vary considerably from those calculated using the constant rate.
- To determine the *P* value for future cash flow values (F_t) at different *i* values (i_t) for each year *t*, we will assume *annual compounding.*
- Define

  i_t = effective annual interest rate for year *t* (*t* = years 1 to *n*)

- To determine the present worth, calculate the P of each F_t value, using the applicable i_t, and sum the results. Using standard notation and the P/F factor,

$$P = F_1(P/F,i_1,1) + F_2(P/F,i_1,1)(P/F,i_2,1) + \cdots$$
$$+ F_n(P/F,i_1,1)(P/F,i_2,1)\cdots(P/F,i_n,1)$$

- The future worth F is determined in a similar fashion. Calculate the F for each P_t value using the applicable it and sum the results to obtain the total F in year n.
- When only single amounts are involved, that is, one P and one F in the final year n, the last term in the Equation is the expression for the present worth of the future cash flow.

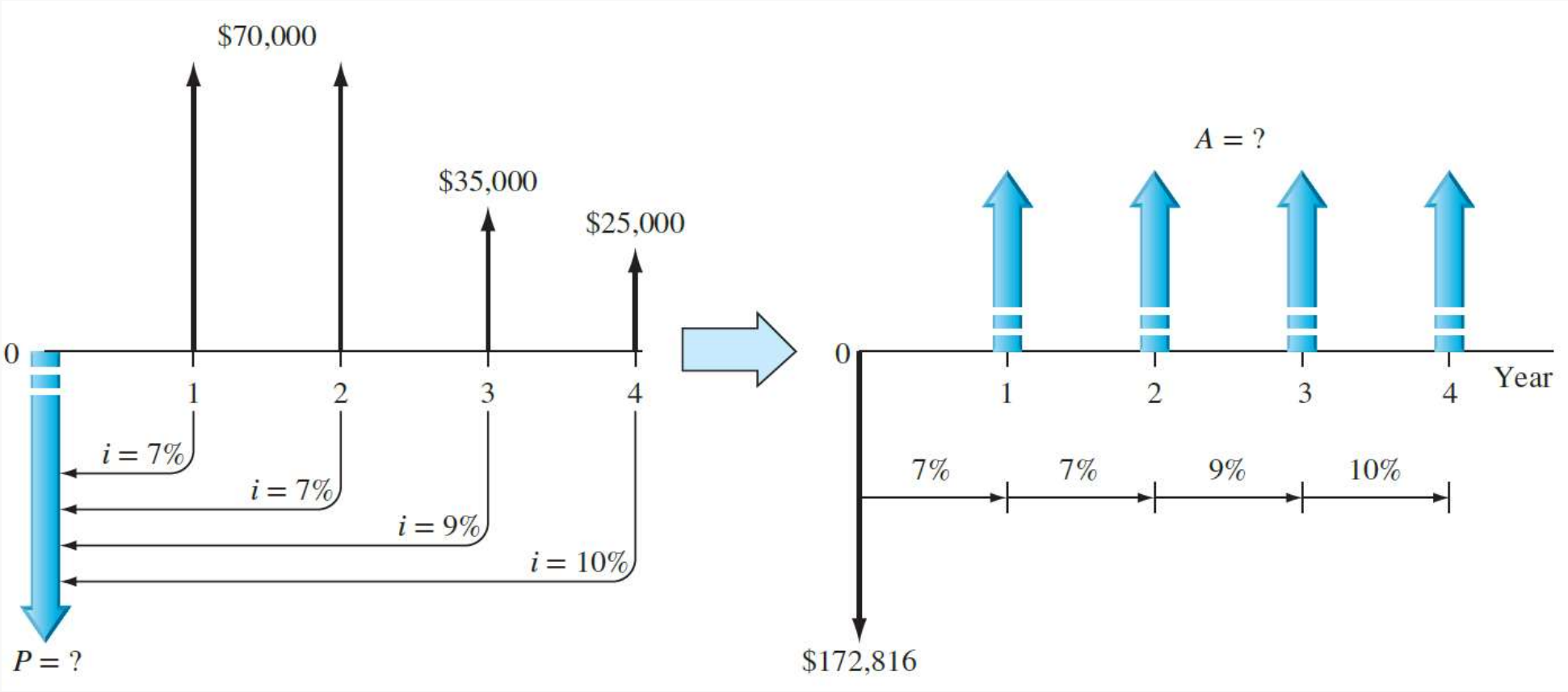
$$P = F_n(P/F,i_1,1)(P/F,i_2,1)\cdots(P/F,i_n,1)$$

- If the equivalent uniform series *A* over all *n* years is needed, first find *P*, using either of the last two equations; then substitute the symbol *A* for each F_t symbol.
- Since the equivalent *P* has been determined numerically using the varying rates, this new equation will have only one unknown, namely, *A*.

EXAMPLE:

- CE, Inc. leases large earth tunneling equipment. The net profit from the equipment for each of the last 4 years has been decreasing, as shown below. Also shown are the annual rates of return on invested capital. The return has been increasing. Determine the present worth *P* and equivalent uniform series *A* of the net profit series. Take the annual variation of rates of return into account.

Year	1	2	3	4
Net Profit	$70,000	$70,000	$35,000	$25,000
Annual Rate	7%	7%	9%	10%



- Since for both years 1 and 2, the net profit is $70,000 and the annual rate is 7%, the *P/A* factor can be used for these 2 years only.

$$P = [70(P/A,7\%,2) + 35(P/F,7\%,2)(P/F,9\%,1)$$
$$+ 25(P/F,7\%,2)(P/F,9\%,1)(P/F,10\%,1)](1000)$$
$$= [70(1.8080) + 35(0.8013) + 25(0.7284)](1000)$$
$$= \$172,816$$

- To determine an equivalent annual series, substitute the symbol *A* for all net profit values on the right side of this Equation, set it equal to *P* = $172,816, and solve for *A*.

$$\$172,816 = A[(1.8080) + (0.8013) + (0.7284)] = A[3.3377]$$
$$A = \$51,777 \text{ per year}$$

- Year 1 PW

$$\frac{70{,}000}{1.07} = 65{,}420.56$$

- Year 2 PW

$$\frac{70{,}000}{1.07^2} = 61{,}140.71$$

- Year 3 PW

Denominator $= 1.07^2 \times 1.09$

$$\frac{35{,}000}{1.07^2 \cdot 1.09} \approx 28{,}036.20$$

- Year 4 PW

Denominator $= 1.07^2 \times 1.09 \times 1.10$

$$\frac{25{,}000}{1.07^2 \cdot 1.09 \cdot 1.10} \approx 18{,}221.82$$

Now sum:

$$P = 65{,}420.56 + 61{,}140.71 + 28{,}036.20 + 18{,}221.82$$

$$\approx \boxed{172{,}819 \text{ (present worth)}}$$

PW of A:

$$P_A = A\left[\frac{1}{1.07} + \frac{1}{1.07^2} + \frac{1}{1.07^2 \cdot 1.09} + \frac{1}{1.07^2 \cdot 1.09 \cdot 1.10}\right]$$

Compute the bracketed term:

$$\frac{1}{1.07} = 0.93458$$

$$\frac{1}{1.07^2} = 0.87344$$

$$\frac{1}{1.07^2 \cdot 1.09} = 0.80132$$

$$\frac{1}{1.07^2 \cdot 1.09 \cdot 1.10} = 0.72847$$

Sum:

$$0.93458 + 0.87344 + 0.80132 + 0.72847 \approx 3.33781$$

Set $P_A = P$:

$$172{,}819 = A \times 3.33781 \Rightarrow A = \frac{172{,}819}{3.33781}$$

$$\approx \boxed{51{,}800 \text{ per year } (\approx \$51{,}776)}$$

- **Comment**
- If the average of the four annual rates, that is, 8.25%, is used, the result is *A* = $52,467.
- This is a $690 per year overestimate of the equivalent annual net profit.

- When there is a cash flow in year 0 and interest rates vary annually, this cash flow must be included to determine *P*.
- In the computation for the equivalent uniform series *A* over all years, including year 0, it is important to include this initial cash flow at $t = 0$.
- This is accomplished by inserting the factor value for $(P/F,i_0,0)$ into the relation for *A*.
- This factor value is always 1.00.
- It is equally correct to find the *A* value using a future worth relation for *F* in year *n*.
- In this case, the *A* value is determined using the *F/P* factor, and the cash flow in year *n* is accounted for by including the factor $(F/P,i_n,0) = 1.00$.

CHAPTER 5:
PRESENT WORTH ANALYSIS

- A future amount of money converted to its equivalent value now has a present worth (PW) that is always less than that of the future cash flow because all P/F factors have a value less than 1.0 for any interest rate greater than zero.
- For this reason, present worth values are often referred to as discounted cash flows (DCF), and the interest rate is referred to as the discount rate. Besides PW, two other terms frequently used are present value (PV) and net present value (NPV).
- Up to this point, PW computations have been made for one project or alternative.
- In this chapter, techniques for comparing two or more mutually exclusive alternatives by the present worth method are treated.

Formulating Alternatives

- Economic evaluation of an alternative requires cash flow estimates over a stated time period and a criterion for selecting the best alternative
- The alternatives are developed from project proposals to accomplish a stated purpose
- Some projects can be economically or technologically viable, some are not
- After the viable projects stand out among the others, these are the alternatives to evaluate

- The nature of the economic proposals is always one of two types:
- **Mutually exclusive alternatives:** Only one of the proposals can be selected. For terminology purposes, each viable proposal is called an *alternative.*
- **Independent projects:** More than one proposal can be selected. Each viable proposal is called a *project.*
- **do-nothing (DN)** option can be another alternative; no new costs or revenues.
- The DN alternative or project means that the **current approach is maintained;** nothing new is initiated. No new costs, revenues, or savings are generated.

What is an independent project?

- Projects are **independent** if:
- Choosing one **does not** affect your ability to choose the others.
- They do not compete for the same budget, resources, or function.
- You can accept **any number** of them.

Examples:

- Installing solar panels at multiple factory locations
- Buying additional machines when the budget is large enough
- Approving R&D projects that do not depend on each other
- Because the projects don’t interact, you evaluate each one **on its own**.

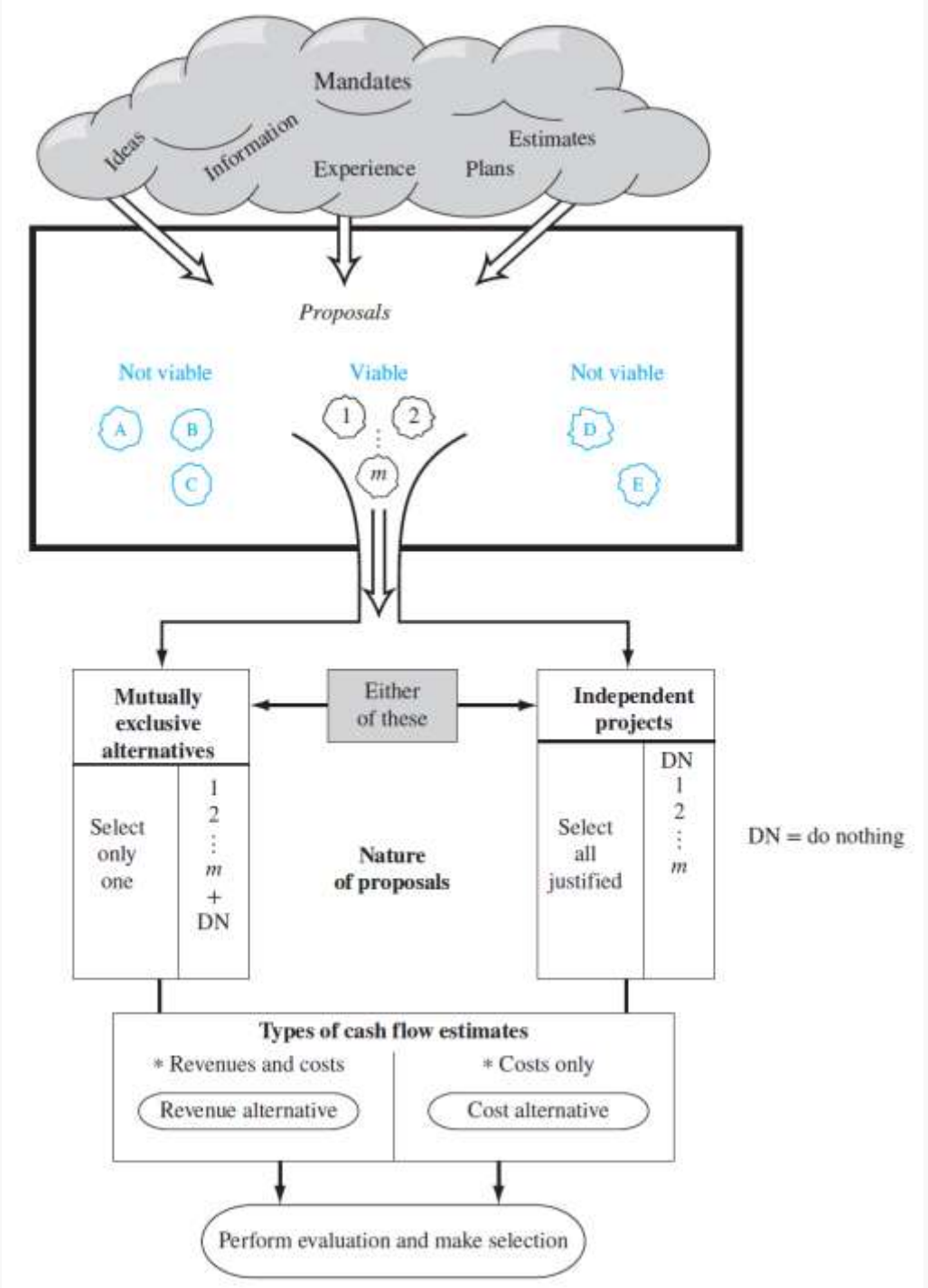
- **Mutually exclusive** alternatives compete with each other, best alternative is selected
- Independent projects compete with the do-nothing alternative instead of another project
- If there are *m* independent projects; one, two or more of them can be selected. (2^m selections!)
- Let we have $m = 3$, and three independent alternatives are
- {*A*, *B*,*C*}
- Then we have $2^3 = 8$ possible selections:
- {*DN*, *A*, *B*,*C*, *AB*, *AC*, *BC*, *ABC*}
- Assumption: No budget limit!

- It is important to recognize the nature or type of alternatives before starting evaluation
- Cash flows determine whether the alternatives are revenue-based or service-based

  1)Revenue-Based: Each alternative generates cost and revenue cash flow estimates and possibly savings

  2)Service-Based: Each alternative has only cost cash flow estimates

    - Revenues or savings are not dependent on alternatives
    - Revenues or savings are assumed to be equal among alternatives
    - Mostly public (government) initiatives
    - Evaluation on based on cost estimates
- All alternatives evaluated in an engineering economy study must be of the same type


Mandates
Ideas
Information
Experience
Estimates
Plans
Proposals
Not viable
Viable
Not viable
A
B
C
1
2
m
D
E
Mutually exclusive alternatives
Either of these
Independent projects
Select only one
1
2
m
+
DN
Nature of proposals
Select all justified
DN
1
2
m
DN = do nothing
Types of cash flow estimates
* Revenues and costs
* Costs only
Revenue alternative
Cost alternative
Perform evaluation and make selection

Present Worth Analysis of Equal-Life Alternatives

- The PW comparison of alternatives with equal lives is straightforward. The present worth *P* is renamed PW of the alternative.
- The present worth method is quite popular in industry because all future costs and revenues are transformed to **equivalent monetary units NOW;** that is, all future cash flows are converted (discounted) to present amounts (e.g., dollars) at a specific rate of return, which is the MARR.
- This makes it very simple to determine which alternative has the best economic advantage.
- **MARR** stands for:
- **Minimum Acceptable Rate of Return**
- MARR is the **minimum return** that a company or decision-maker requires before accepting a project or an investment alternative.

- Basic procedure for applying the Present Worth (PW) criterion:

  1)Determine the interest rate (*i*) that the firm wishes to earn on its investments. *i* = *MARR* is usually given to us.

  2)Estimate the service life (*n*) of the project

  3)Estimate the cash inflow for each period over the service life

  4)Estimate the cash outflow for each period over the service life

  5)Determine the net cash flows (*An*) for each period (net cash flow = cash inflow - cash outflow)

  6)Find the present worth of each net cash flow at MARR. Add up these present-worth figures; their sum is defined as the projects

- *PW*:

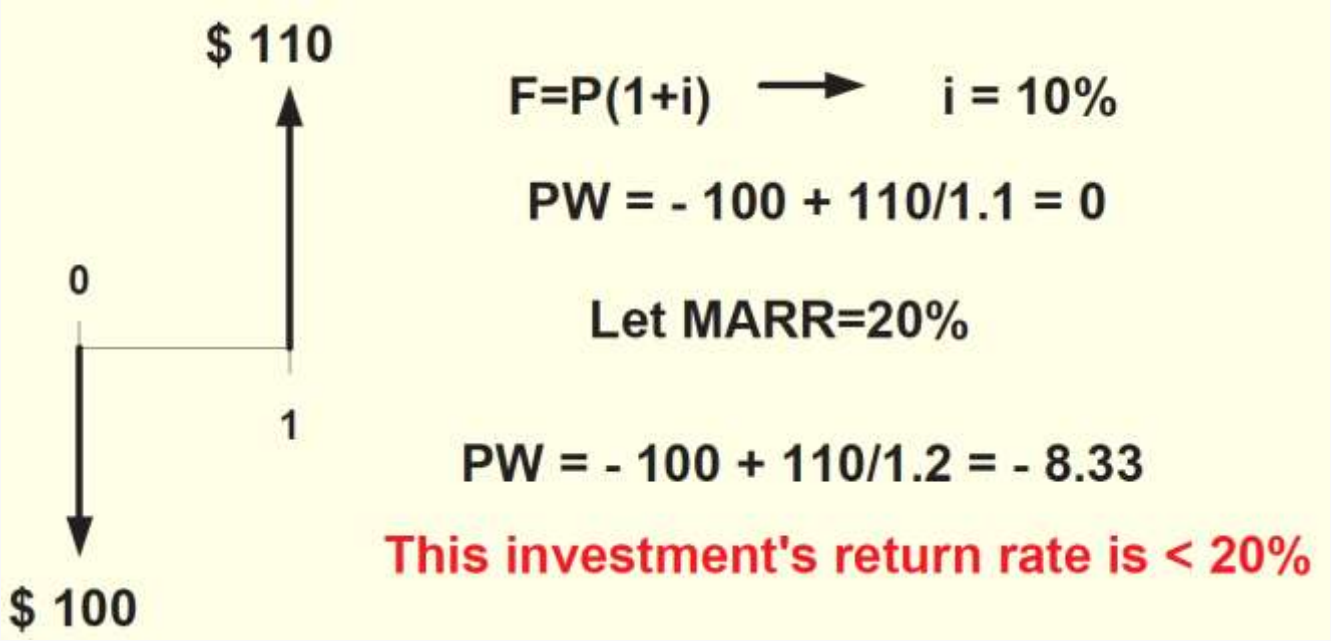
$$PW(i) = \frac{A_0}{(1+i)^0} + \frac{A_1}{(1+i)^1} + \cdots + \frac{A_N}{(1+i)^N}$$

$$= \sum_{n=0}^{N} \frac{A_n}{(1+i)^n} = \sum_{n=0}^{N} A_n(P/F, i, n)$$

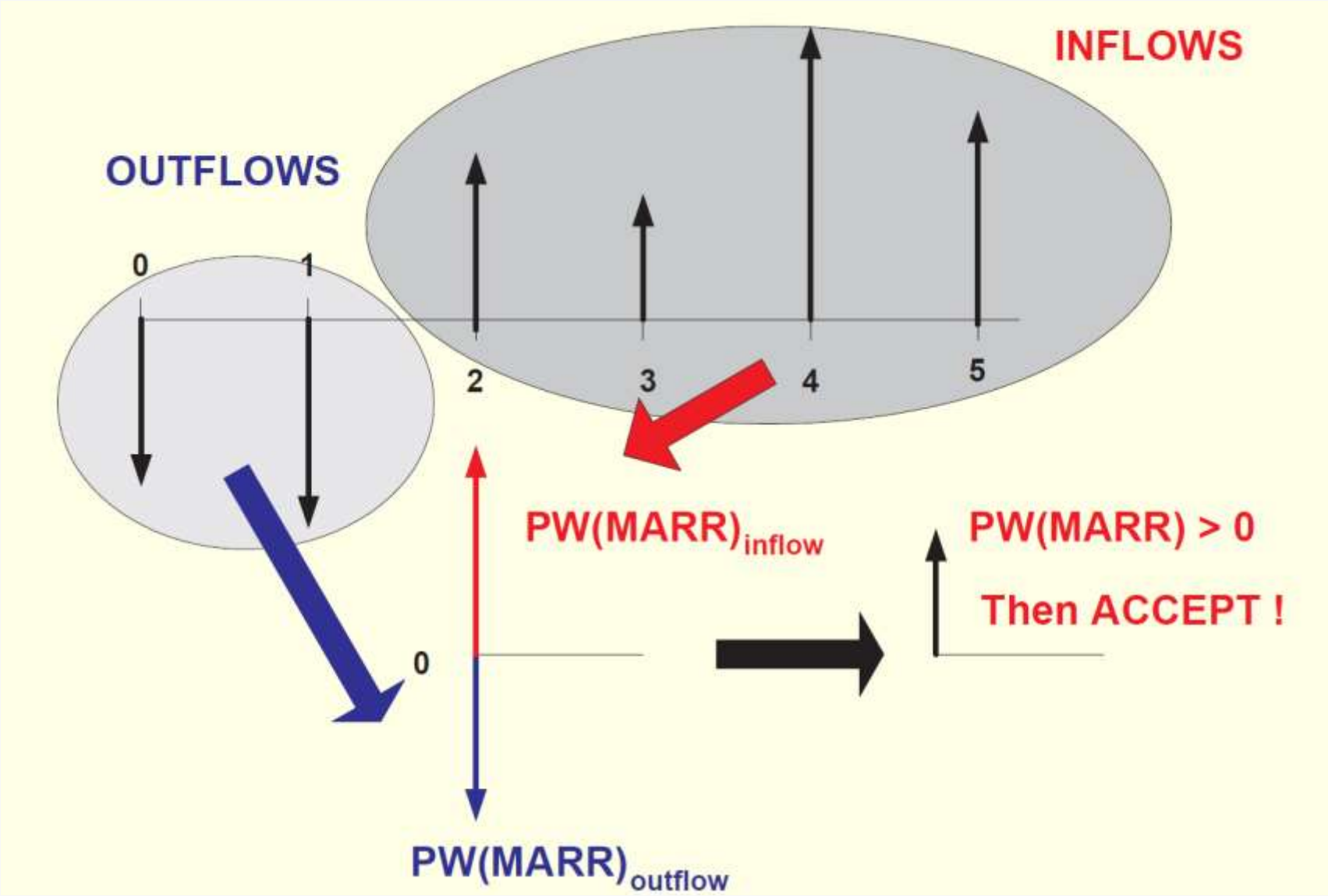
- *P* value or now called Present Worth *PW* is calculated using MARR for each alternative.
- When the alternatives have equal lives, while carrying out the *PW* analysis for MUTUALLY EXCLUSIVE ALTERNATIVES:

  1) One Alternative: Calculate *PW* at MARR. If *PW* 0, the requested MARR is met or exceeded and the alternative is financially viable.

  2) Two or more Alternatives: Calculate *PW* of each alternative at MARR. Select the alternative with the *PW* value that is numerically largest. (not absolute value)



- *PW* analysis is popular because future costs and revenue estimates are all transformed into equivalent dollars now; all future cash flows are converted into present dollars
- Makes easier to determine the economic advantage of one alternative over another



- Note that the guideline to select one alternative with the lowest cost or highest revenue uses the criterion of **numerically largest.**
- This is not the absolute value of the PW amount because the sign matters. The selections below correctly apply the guideline for two alternatives 1 and 2.

Example

PW_1	PW_2	Selected One
$-1500	$-500	2
-500	1000	2
2500	-500	1
2500	1500	1

- If the projects are INDEPENDENT (and revenue projects), the selection guideline is as follows:
- For one or more independent projects, select all projects with $\boldsymbol{PW \geq 0}$ at the MARR
- Why?
- A project with **PW ≥ 0** earns **at least the Minimum Acceptable Rate of Return (MARR).**
- That means the project is profitable at your required return level.
- Since projects are independent, choosing one does **not limit** choosing others.
- Therefore, the best decision is simply:

☞ **Accept every project that meets your minimum required return.**

- **Why PW ≥ 0 means "economically good"?**
- PW converts all future cash flows into **present value terms**:

$$PW = \sum \frac{CF_t}{(1+i)^t}$$

If PW ≥ 0:

- The project **covers its costs**, including opportunity cost of capital.
- It earns at least **i = MARR**.
- Your money grows at or above the required rate.

If PW < 0:

- The project earns **less than the MARR**.
- It is economically inferior to simply investing at the MARR.

EXAMPLE

- MARR = 12%

Project A:

- Initial cost: –$50,000
- Future benefits: $70,000 (after 1 year)

$$PW = -50,000 + \frac{70,000}{1.12} = +12,500 > 0$$

→ **Accept Project A**

EXAMPLE

Project B:

- Initial cost: –$40,000
- Future benefits: $40,000 (after 1 year)

$$PW = -40,000 + \frac{40,000}{1.12} = -3,571 < 0$$

→ **Reject Project B**

Example

PW_1	PW_2	Selected One
$-1500	$-500	DN
-500	1000	2
2500	-500	1
2500	1500	1,2

- Summary
- ✓ Independent projects do not interact
- ✓ Evaluate each project on its own
- ✓ Accept all projects with PW ≥ 0 at the MARR
- ✓ Reject any project with PW < 0

Example

Perform a present worth analysis of equal-service machines with the costs shown below, if the MARR is 10% per year. Revenues for all three alternatives are expected to be the same.

	Electric-Powered	Gas-Powered	Solar-Powered
First cost,$	$-2500	$-3500	-6000
Annual Operating cost, $	-900	-700	-50
Salvage value, $	200	350	100
Life, years	5	5	5

- Service alternatives
- Salvage value considered as a negative cost, so + sign used
- *PW* of each alternative is calculated at i = 10%, for n = 5 years.

Example

	Electric-Powered	Gas-Powered	Solar-Powered
First cost,$	$-2500	$-3500	-6000
Annual Operating cost, $	-900	-700	-50
Salvage value, $	200	350	100
Life, years	5	5	5

$$
\begin{aligned}
PW_E &= -2500 - 900(P/A, 10\%, 5) + 200(P/F, 10\%, 5) \\
&= -2500 - 900(3,7908) + 200(0.6209) = \$ - 5787.54 \\
PW_G &= -3500 - 700(P/A, 10\%, 5) + 350(P/F, 10\%, 5) \\
&= -3500 - 700(3,7908) + 350(0.6209) = \$ - 5936.245 \\
PW_S &= -6000 - 50(P/A, 10\%, 5) + 100(P/F, 10\%, 5) \\
&= -6000 - 50(3,7908) + 100(0.6209) = \$ - 6127.45
\end{aligned}
$$

Select Electric-powered machine with $PW_E = \$ - 5787.54$

- If you want to compare **mutually exclusive alternatives** (you can choose only one), you need to consider that **they have different service lives.**

Example:

- Alternative A lasts **4 years**
- Alternative B lasts **6 years**

If you compare their PW values **directly**, it is *not fair* because:

- A gives service for 4 years
- B gives service for 6 years
- You must make sure you are comparing **equal amounts of service.**
- So, you need to **match** the lifetimes
- This is called the **equal-service requirement**.

- When the present worth method is used to compare MUTUALLY EXCLUSIVE alternatives that have different lives, the procedure is same as before except:
  - *PW* of alternatives must be compared over the same number of years and end at the same time
  - A fair comparison can be made only when PW values represent costs (and receipts) associated with equal service
  - The equal-service requirement can be satisfied by either of two approaches:
    1) LCM Method: Compare the alternatives over a period of time equal to least common multiple (LCM) of their lives
    2) Planning Horizon Method: Compare the alternatives using a study period of length of *n* years, which does not necessarily take into consideration the useful lives of the alternatives.
- In either case, MARR is used to calculate the PW of each alternative

- The LCM approach makes the cash flows for all alternatives extend to the same time period.

Example:

- A lasts 4 years
- B lasts 6 years
- LCM = 12 years

So:

- Repeat A ⟶ 3 cycles (4 + 4 + 4)
- Repeat B ⟶ 2 cycles (6 + 6)
- You compute PW over **12 years** for each alternative.
- This method assumes:

☞ the alternatives can be repeated indefinitely

☞ their cost/benefit pattern repeats in the same way

- In either case, MARR is used to calculate the PW of each alternative
- The LCM approach makes the cash flows for all alternatives extend to the same time period
- The LCM approach requires some assumptions to be made about subsequent life cycles of the alternatives:
    1) The service provided by alternatives will be needed for the LCM of years or more
    2) The selected alternative will be repeated over each life cycle of the LCM in exactly the same manner
    3) The cash flow estimates will be the same in every life cycle
- The LCM requires that the estimated salvage values be included in each life cycle

- In Planning Horizon Method (n-Year Study Period) instead of repeating alternatives, you simply choose a study period, say **n years**:
  - maybe 10 years (company planning horizon)
  - maybe the life of the longest alternative
  - maybe any convenient number
- You then compute PW for each alternative **over the same study period**.

- What if an alternative lasts longer or shorter than the horizon?
- If its life is **shorter**, you assume you must replace it (replacement costs added).
- If its life is **longer**, you assume a **salvage value** at the end of year n.
- This method does **not** assume repeatability.
- Also useful when LCM of alternatives yields unrealistically long time horizons

Eric Forman, a project engineer is assigned to start up a new office in Wisconsin. Two lease options are available:

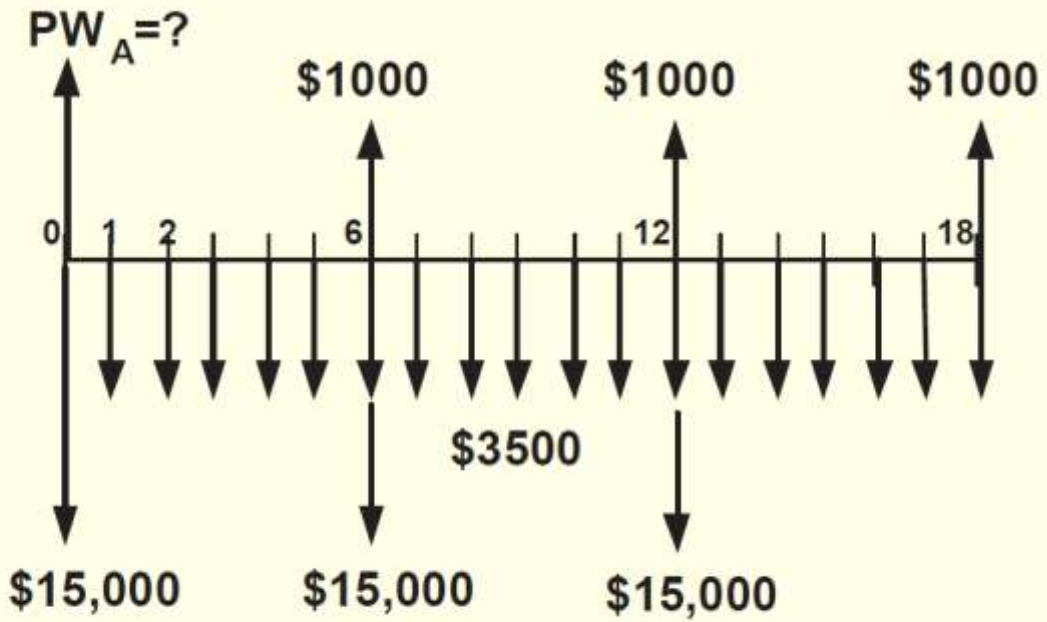
	Location A	Location B
First cost, $	$-15,000	$-18,000
Annual lease cost, $ per year	-3,500	-3,100
Deposit return, $	1,000	2,000
Lease term, years	6	9

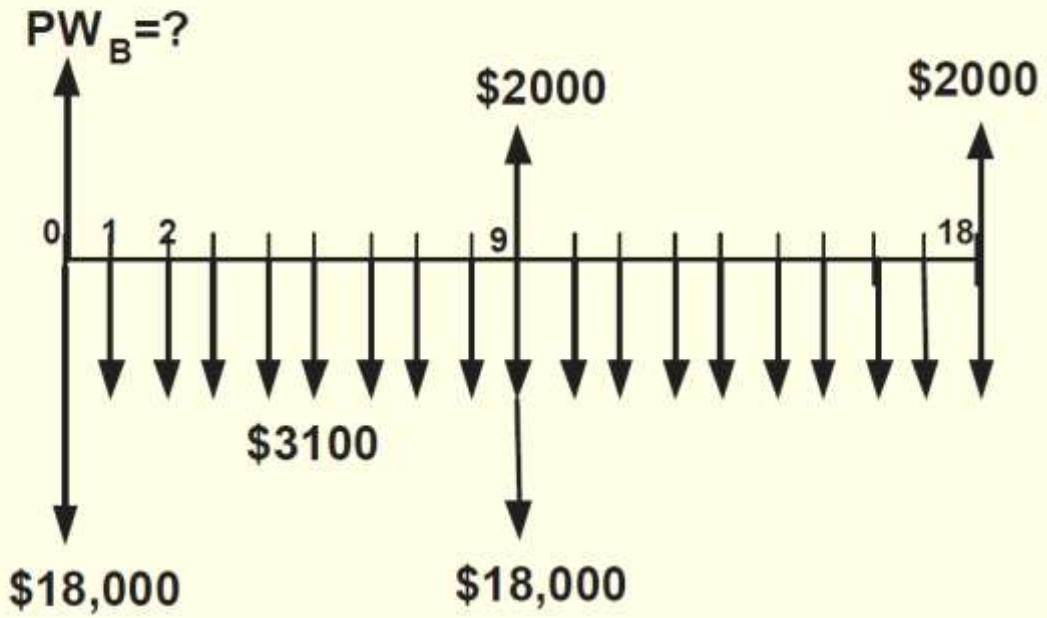
a) Determine which lease option should be selected on the basis of a present worth comparison if MARR is 10% per year.

b) If a study period of 5 years is used and the deposit returns are not expected to change, which location should be selected

c) Which location should be selected if a 6-year study period and a deposit return of $6,000 after 6 years at location B is used.

a)


PW A=?
$1000
$1000
$1000
0
1
2
6
12
18
$3500
$15,000
$15,000
$15,000


PW B=?
$2000
$2000
0
1
2
9
18
$3100
$18,000
$18,000

$$
\begin{aligned}
PW_A &= -15,000 - 3,500(P/A, 10\%, 18) + 1,000(P/F, 10\%, 6) \\
&- 15,000(P/F, 10\%, 6) - 15,000(P/F, 10\%, 12) \\
&+ 1,000(P/F, 10\%, 12) + 1,000(P/F, 10\%, 18)
\end{aligned}
$$

$$
\begin{aligned}
PW_A &= -15,000 - 3,500(8.2014) + 1,000(0.5645) \\
&- 15,000(0.5645) - 15,000(0.3186) + 1,000(0.3186) \\
&+ 1,000(0.1799) = \$ - 55,888.4
\end{aligned}
$$

$$
\begin{aligned}
PW_B &= -18,000 - 3,100(P/A, 10\%, 18) + 2,000(P/F, 10\%, 9) \\
&- 18,000(P/F, 10\%, 9) + 2,000(P/F, 10\%, 18) \\
&= -18,000 - 3,100(8.2014) + 2,000(0.4241) \\
&- 18,000(0.4241) + 2,000(0.1799) = \$ - 49,850
\end{aligned}
$$

Location B is selected.

b) For a 5-year study period no cycle repeats are necessary.

$$\begin{aligned} PW_A &= -15,000 - 3,500(P/A, 10\%, 5) + 1,000(P/F, 10\%, 5) \\ &= -15,000 - 3,500(3,7908) + 1,000(0,6209) \\ &= \$ - 27,647 \\ PW_B &= -18,000 - 3,100(P/A, 10\%, 5) + 2,000(P/F, 10\%, 5) \\ &= -18,000 - 3,100(3,7908) + 2,000(0,6209) \\ &= \$ - 28,509.7 \end{aligned}$$

Location A is selected.

c) For a 6-year study period, deposit return of $6,000 for location B

$$\begin{aligned} PW_A &= -15,000 - 3,500(P/A, 10\%, 6) + 1,000(P/F, 10\%, 6) \\ &= -15,000 - 3,500(4.3553) + 1,000(0,5645) \\ &= \$ - 29,679 \\ PW_B &= -18,000 - 3,100(P/A, 10\%, 6) + 6,000(P/F, 10\%, 6) \\ &= -18,000 - 3,100(4.3553) + 6,000(0.5645) \\ &= \$ - 28,114 \end{aligned}$$

Location B is selected.

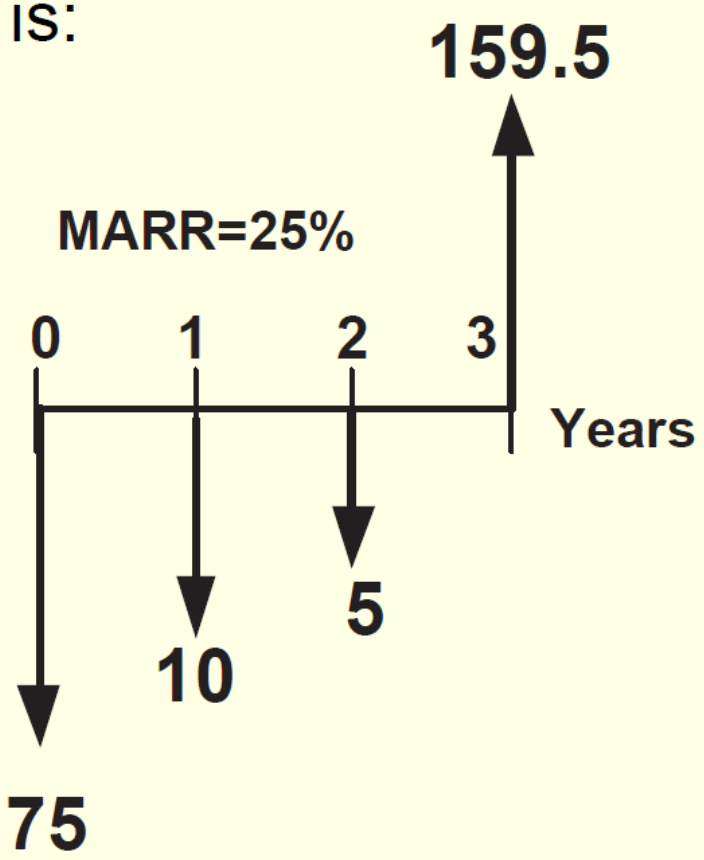
- Future Worth (*FW*) of an alternative can be found:
  1)Directly from the cash flows by determining the future worth value
  2)multiplying *PW* value by the *F/P* factor at
- MARR
- Therefore it is an extension of *PW* analysis
- Used when the prime goal is to maximize the future wealth of the company
- Also used if the asset will be sold or traded before its expected life is reached
  - *FW* value estimates the alternative's worth at the time sale or disposal
- Selection guidelines are same as *PW*
  1)One alternative: *FW* 0 means MARR is met
  2)For two or more mutually exclusive: Select the numerically largest

Example

A British food distribution corporation purchased a Canadian food store chain for 75 million dollars three years ago. There was a net loss of $10 million at the end of year 1 of ownership. Net cash flow is increasing with an arithmetic gradient of $+5 million per year starting the second year. This means that breakeven net cash flow was achieved this year. Because of the heavy debt financing used to purchase the Canadian chain, the international board of directors expects a MARR of 25% per year from any sale.

1. The British corporation has just been offered $159.5 million by a French company for the ownership of the Canadian chain. Use *FW* analysis to decide whether the MARR is met at this selling price or not.
2. If the British corporation continues to own the chain, what selling price must be obtained at the end of the year 5 to make the MARR.

1. Cash flow diagram is:

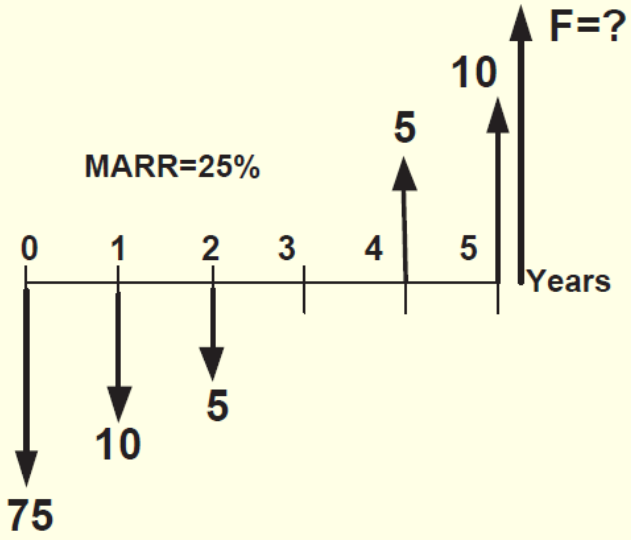


$$FW_3 = -75(1.25)^3 - 10(1.25)^2 - 5(1.25) + 159.5$$

$$= -8.86$$

<0, MARR of 25% not met

1. **<0, MARR of 25% not met**
2. cash flow diagram is:



$$FW_3 = -75(1.25)^5 - 10(1.25)^4 - 5(1.25)^3 + 5(1.25) + 10 + F$$

$$0 = F - 246.81$$

$$F = \$246.81 \textit{ million}$$

at least to meet MARR of 25%

- Capitalized Cost (CC) is the present worth of an alternative that will last forever
- Public sector projects such as bridges, dams, railroads fall into this category
- CC is derived from $P = A(P/A, i\%, n)$ as $n \to \infty$

$$\begin{aligned} P &= A\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right] \\ &= A\left[\frac{1 - \frac{1}{(1+i)^n}}{i}\right] \end{aligned}$$

- As $n \to \infty$, the expression inside the bracket becomes $1/i$, and the symbol CC replaces P
- $CC = \frac{A}{i} = \frac{AW}{i}$

Capitalized Cost (CC) method

- **Capitalized Cost (CC) method**, used when a project continues **forever** (infinite life), such as:
- roads
- bridges
- public buildings
- long-term maintenance systems
- campus facilities
- Capitalized Cost answers:
- **What is the lump-sum amount needed today to fund this project forever?**

- To compute CC, you must treat different types of costs correctly:
- **Recurring costs** → occur every cycle (every N years)
- **Nonrecurring costs** → occur only once (or irregularly)
- The following procedure assists in calculating the CC for an infinite sequence of cash flows.

  1) Draw a cash flow diagram for all nonrecurring cash flows and at least two cycles of all recurring cash flows
  2) Find the present worth of all nonrecurring amounts. This will be their *CC* value
  3) Find the equivalent uniform annual worth (*A*) through one life cycle of all recurring amounts. Add this to all other uniform amounts occurring in years 1 through infinity and the result is total equivalent uniform annual worth (*AW*)
  4) Divide the *AW* obtained at step 3 by *i* to obtain a *CC* value
  5) Add the *CC* values obtained in steps 2 and 4

Step 1: Draw a cash flow diagram

- You draw:
- All **nonrecurring costs** (e.g., construction cost at year 0, a major overhaul at year 15).
- At least **two cycles** of all recurring costs (e.g., maintenance cost every 4 years).
- Why two cycles?

→ To clearly identify the repeating pattern.

- This helps you visually separate:
- costs that repeat
- costs that happen only once

Step 2: Present worth of all nonrecurring amounts

- All cash flows that occur **only once** (or only at irregular times) must be converted to **present value now**.
- This gives the **CC for nonrecurring costs**.

- Example:
  Initial construction: –$5,000,000
  A major repair at year 20: –$400,000
- You discount each back to time 0.

- **Step 3: Find the uniform annual worth (AW) of recurring costs**
- Recurring costs repeat every cycle.
- For example:

  repaint every 6 years: –$80,000

  resurface every 12 years: –$200,000
- You:
- Convert each repeating cycle cost into an **Equivalent Uniform Annual Cost (EUAC or AW)** for *one* cycle.
- Add all recurring annual-equivalent costs → **total AW** for recurring items.
- This step transforms periodic costs into a smooth **annual cost**.

- **Step 4: Convert annual worth (AW) to capitalized cost**
- Since the project lasts **forever**, you convert AW to a capitalized cost using:

$$CC = \frac{AW}{i}$$

- Why divide by i?

- Because for a perpetual uniform series:

$$PW = \frac{A}{i}$$

- This gives the **present worth of all recurring costs forever**.

- **Step 5: Add CC of nonrecurring and recurring costs**
- Final capitalized cost:

$$CC_{total} = CC_{nonrecurring} + CC_{recurring}$$

- This is the **total money needed today** to fund the entire project forever.

EXAMPLE

- Suppose:
- Construction cost today: –$1,000,000 (nonrecurring)
- Maintenance every 5 years: –$50,000 (recurring)
- Interest rate i=5%
- **Step 2:**
  Nonrecurring CC = –$1,000,000.
- **Step 3:**
  Convert recurring cost to annual:
- AW=50,000(A/P,5%,5)

- **Step 4:**
  Capitalized cost of recurring cost:

$$CC_{\text{recurring}} = \frac{AW}{i}$$

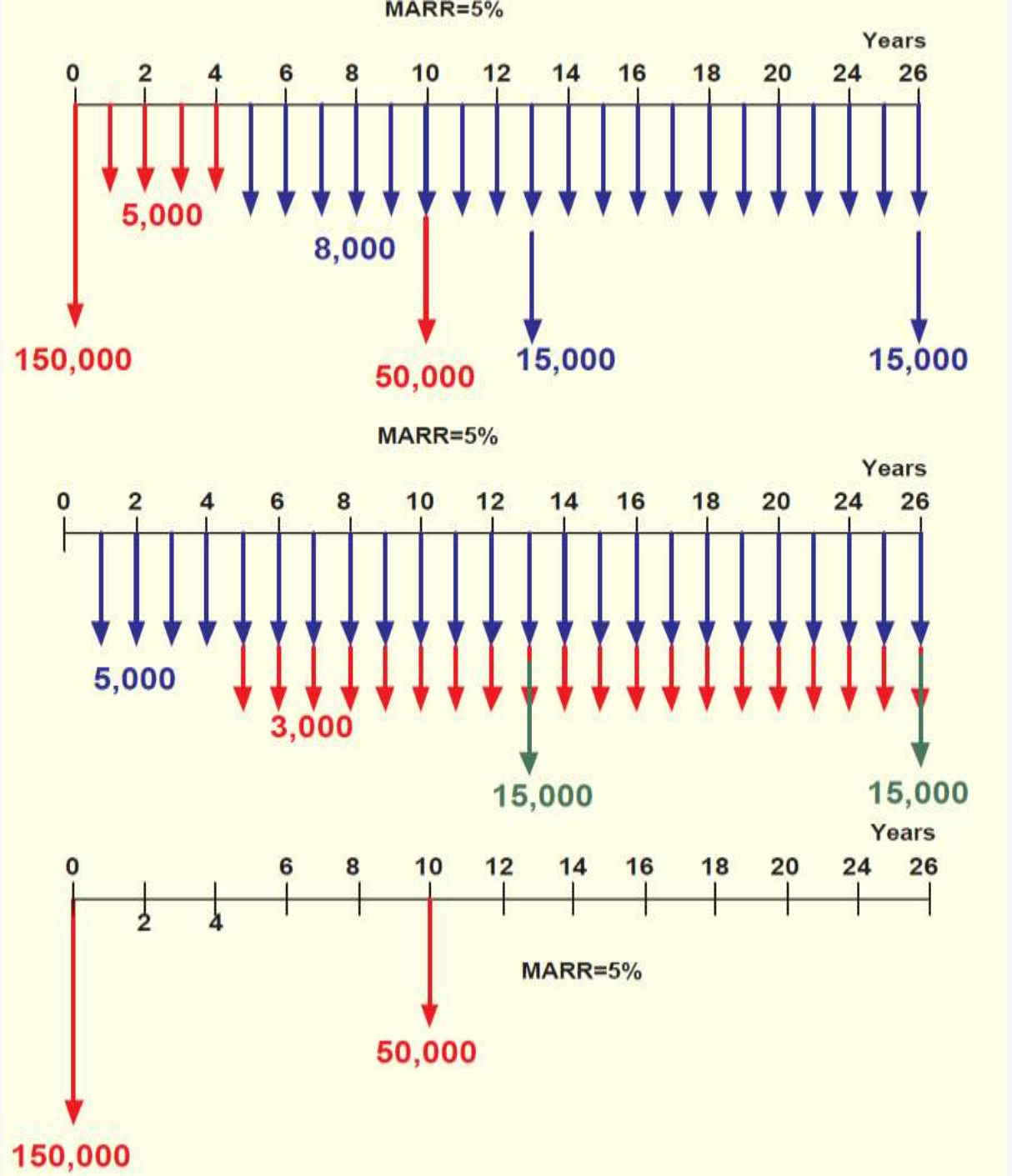
- **Step 5:**

- Add both parts:
- $CC_{total} = 1{,}000{,}000 + CC_{recurring}$

- That gives the total money required **today** to support the project forever.

A new software for the purpose of monitoring property tax is installed. The installation cost is $150,000 and there is an additional cost of $50,000 after 10 years. The annual software contract cost is $5,000 for the first 4 years and $8,000 thereafter. In addition, there is expected to be a recurring major upgrade cost of $15,000 every 13 years. Assume that MARR is 5%. If the new software will be used for the indefinite future, find the equivalent value:

1. now
2. for each year thereafter


MARR=5%
Years
0 2 4 6 8 10 12 14 16 18 20 24 26
5,000
8,000
150,000
50,000
15,000
15,000
MARR=5%
Years
0 2 4 6 8 10 12 14 16 18 20 24 26
5,000
3,000
15,000
15,000
Years
0 2 4 6 8 10 12 14 16 18 20 24 26
MARR=5%
50,000
150,000

2)

$$CC_1 = -150,000 - 50,000\left(\frac{1}{(1.05)^{10}}\right)$$
$$= \$-180,695$$

3)

$$A_1 = -15,000(A/F, 5\%, 13)$$
$$= -15,000(0.05646) = \$-847$$

4)

$$CC_2 = \frac{-3,000}{0.05}\left(\frac{1}{1.05^4}\right)$$
$$= \$-49,362$$

$$CC_3 = A_1 + A_2 = \frac{-5,000 - 847}{0.05} = \$-116,940$$

5) $CC_T = -180,695 - 49,362 - 116,940 = \$-346,997$

b) $A = Pi = -346,997(0.05) = -17,350$

Example

Two alternative sites are considered for a bridge to cross a river in New York.

1. A suspension bridge will cost $50 million with annual inspection and maintenance costs of $35,000. In addition, the concrete deck would have to resurfaced every 10 years at a cost of $100,000. The cost of purchasing right-of-way is expected to be $2 million for the suspension bridge.
2. A truss bridge and approach roads are expected to cost $25 million and annual maintenance costs of $20,000. The bridge have to be painted every 3 years at a cost of $40,000. In addition, the bridge would have to be sandblasted every 10 years at a cost of $190,000. The cost of purchasing right-of-way is expected to be $15 million for the truss bridge.

Compare the alternatives on the basis of their capitalized cost if the interest rate is 6% per year.

- To compare two or more alternatives on the basis of capitalized cost, find CC_T for each alternative.
- CC_T represents the total present worth of financing and maintaining an alternative forever
- Alternative with the smallest capitalized cost should be selected

1. $CC_1 = -50 - 2 = \$ - 52$ million

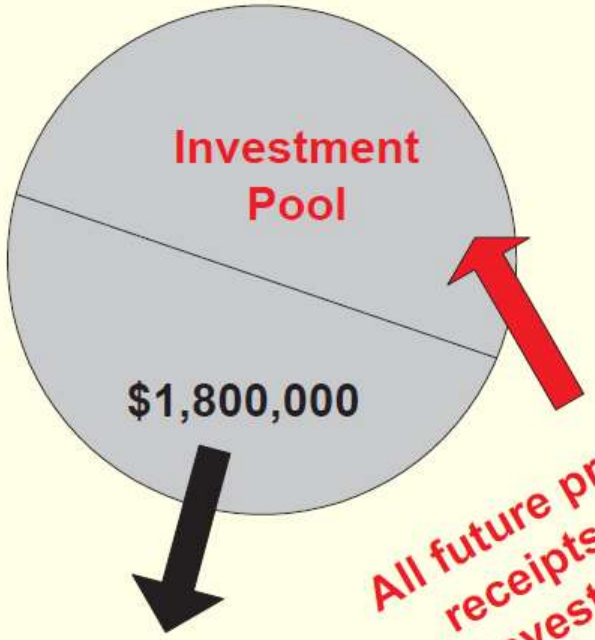
$$
\begin{aligned}
A_1 &= -100,000(A/F, 6\%, 10) \\
&= -100,000(0.07587) = \$ - 7587 \\
CC_2 &= \frac{A_1 + A_2}{i} = \frac{-7,587 - 35,000}{0.06} \\
&= -709,783 \\
CC_S &= CC_1 + CC_2 = \$ - 52,71 \ \textit{million}
\end{aligned}
$$

1. $CC_S = \$ - 52,71$ million
2. $CC_1 = -25 - 15 = \$ - 40$ million

$$
\begin{aligned}
A_1 &= -190,000(A/F, 6\%, 10) \\
&= -190,000(0.07587) = \$ - 14,415 \\
A_2 &= -40,000(A/F, 6\%, 3) \\
&= -40,000(0.31411) = \$ - 12,564 \\
CC_2 &= \frac{A_1 + A_2 + A_3}{i} = \frac{-14,415 - 12,564 - 20,000}{0.06} \\
&= -782,983 \\
CC_T &= CC_1 + CC_2 = \$ - 40,78 \ \textit{million}
\end{aligned}
$$

Truss Bridge is selected.

- If the firm does not spend its money on a project, it can invest that money and earn MARR.
  So every project must earn at least MARR to be worthwhile.



Take 1,800,000 out of the investment pool and invest in the project

All future project cash receipts return to investment pool

How much money would you have if the investment was made?

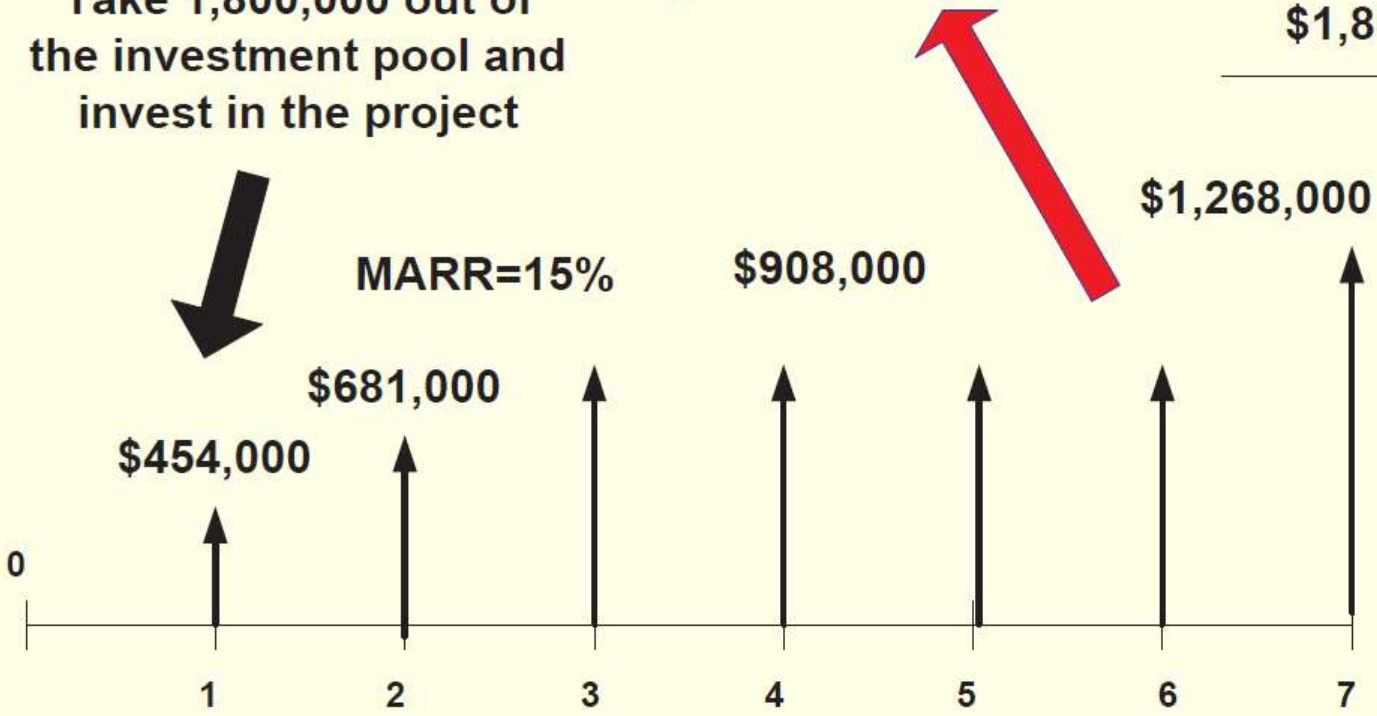
$454,000(F/P, 15%, 6)=$1,050,130
$681,000(F/P, 15%, 5)=$1,369,734
$908,000(F/A,15%,4)(F/P, 15%, 1)=$4,534,007
$1,268,000(F/P,15%,0)= $1,268,000
$8,901,946

How much money would you have if the investment was not made?

$1,800,000(F/P, 15%, 7)=$4,788,036

What is the net gain from investing into the project?

$8,901,946
— $4,788,036
$4,113,910



- Your firm has **$1,800,000** in its investment pool.
- The investment pool earns **15% interest (MARR)**.
- You are evaluating **whether to invest** $1,800,000 in a project.
- The project generates a series of **future cash inflows**:
  - *Year 1: $454,000*
  - *Year 2: $681,000*
  - *Year 3: $908,000*
  - *Year 6: $1,268,000*
    (and similar)
- All **future project receipts** go BACK into the investment pool at 15%.
- The goal:
- Find how much money you would have after 7 years if you invest in the project vs. if you do nothing.
- Then compare the future values.

STEP 1 — Take money out of the investment pool

- At time 0, you withdraw:
- $1,800,000 and use it to start the project.

STEP 2 — The project generates revenues, which return to the pool

- Each cash inflow is deposited into the pool **and earns 15% per year until year 7.**
- For example:
- **Year 1 inflow = $454,000**
- It earns interest for 6 more years (from year 1 to year 7):
- 454,000(F/P,15%,6) = 1,050,130

- **Year 2 inflow = $681,000**
- Earns interest for 5 years:
- 681,000(F/P,15%,5) = 1,369,734
- **Continues similarly for all inflows…**
- All inflows are converted into their value **at year 7** and added together.

STEP 3 — Add all future values when the project IS made

- The diagram shows:
- $8,901,946
- This is the total value of:
  - all project returns
  - grown forward to year 7
  - inside the investment pool.

STEP 4 — What if you DO NOT invest?

- If you leave the **$1,800,000** in the investment pool, it grows at 15% for 7 years:
- 1,800,000(F/P,15%,7) = 4,788,0361
- So by **doing nothing**, you have:
- $4,788,036

STEP 5 — Net Benefit From Investing

- Compare the two amounts:
- 8,901,946 – 4,788,036 = 4,113,910
- This is the **net gain** from choosing the project instead of keeping the money in the pool.

- FINAL INTERPRETATION
- If you invest in the project → You end up with $8.90 million in 7 years.
- If you don’t invest → You end up with $4.79 million in 7 years.
- ✓ Net gain: $4.11 million
- Therefore, the project is **worth doing**, because it earns **more than the investment pool at 15% MARR**.